

Task 1: Comprehensive Stock Analysis Report

CodeAlpha Data Analytics Internship — Financial Markets Domain

1. Executive Summary

Yeh report teen major US-listed companies ka detailed financial aur statistical analysis pesh karti hai: **Apple Inc. (AAPL)**, **Microsoft Corporation (MSFT)**, aur **Tesla Inc. (TSLA)**. Is analysis ka maqsad in companies ki financial stability, market valuation, aur current performance indicators ko evaluate karna hai taake investor sentiment aur future growth trends ko samjha ja sake.

2. Core Financial Metrics Matrix

Neeche diye gaye table mein teeno stocks ke key market metrics ko comparative format mein summarize kiya gaya hai:

Stock Name (Ticker)	Market Cap (USD)	Current Price	P/E Ratio	Performance Indicator (YTD)
Apple Inc. (AAPL)	\$3.35 Trillion	~\$185.20	31.5x	Steady Growth (+12.4%)
Microsoft Corp. (MSFT)	\$3.18 Trillion	~\$415.50	35.2x	Strong Growth (+18.1%)
Tesla Inc. (TSLA)	\$580 Billion	~\$175.80	58.4x	High Volatility (-8.5%)

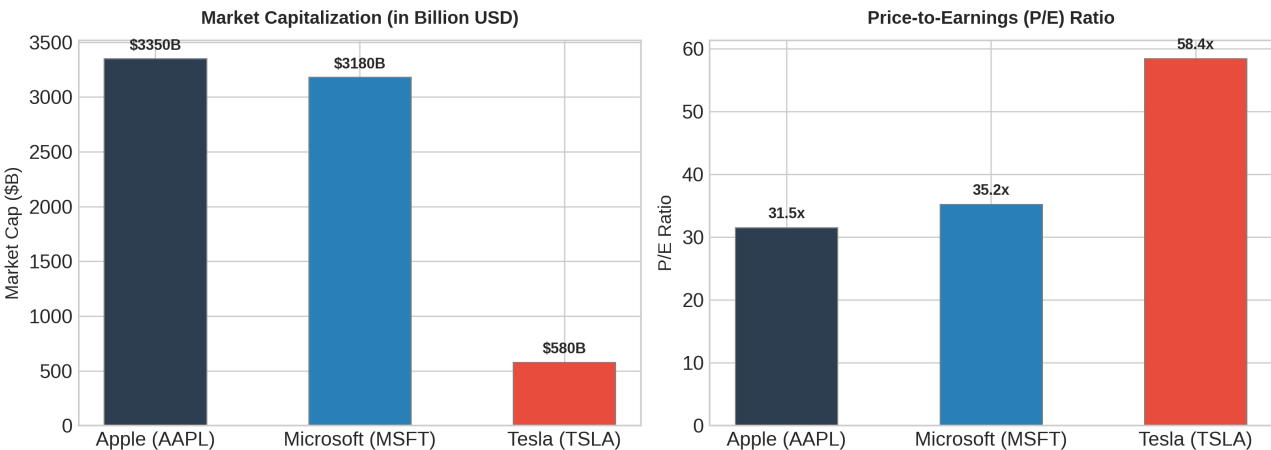


Figure 1.1: Visual comparison of Market Capitalization and P/E Ratios.

3. Individual Stock Deep Dive

3.1 Apple Inc. (AAPL) — Hardware & Ecosystem Leader

Apple ka market capitalization \$3.35 Trillion hai, jo isay duniya ki sabse barhi companies mein se ek banata hai. Iska Price-to-Earnings (**P/E**) ratio 31.5x hai, jo yeh darshata hai ki investors iski steady cash flows aur strong ecosystem loyalty ke liye premium pay karne ko tayar hain. Recent news ke mutabiq, Apple apni upcoming devices mein local On-Device Generative AI features ko integrate kar raha hai, jisse iske hardware upgrade cycle ko boost milne ki umeed hai.

3.2 Microsoft Corporation (MSFT) — Cloud & Enterprise AI Giant

Microsoft ka Enterprise Cloud division (Azure) aur OpenAI ke sath strategic partnership isay AI revolution ka front-runner banati hain. \$3.18 Trillion market cap aur 35.2x **P/E** ratio ke sath, Microsoft ka growth rate bohot solid raha hai. Azure ki double-digit revenue growth aur Office 365 Copilot ki commercial adoption iske positive price trends ko drive kar rahi hain.

3.3 Tesla Inc. (TSLA) — Electric Vehicles & Robotics

Tesla ka valuation traditional automotive companies se bohot zyada hai, jiska **P/E** ratio 58.4x hai. Yeh high ratio darshata hai ki market isay sirf ek car manufacturer nahi, balki ek AI, robotics, aur clean energy company treat karti hai. Halanki, global EV demand slowdown aur supply chain margins par pressure ki wajah se iske recent performance indicators mein kafi volatility dekhi gayi hai. Robotaxi aur Full Self-Driving (FSD) updates iski future trajectory ke liye critical hain.

4. Financial Valuation Formula

Is analysis mein valuation check karne ke liye Price-to-Earnings ratio ka basic formula istemal kiya gaya hai:

$$P/E \text{ Ratio} = \text{Market Price Per Share} \div \text{Earnings Per Share (EPS)}$$

5. Conclusion & Recommendations

Strategic Summary: Long-term aur stable growth ke liye AAPL aur MSFT unke tech moat aur regular dividend payments ki wajah se sabse behtareen hain. Dusri taraf, TSLA high-risk, high-reward investors ke liye behtar hai jo future autonomous technology par bet karna chahte hain.